

COGS25.01 • PHILOSOPHY OF COGNITIVE SCIENCE

Winter 2026

Instructor:	Caleb Kendrick	Time:	MWF 2:10–3:15
Office Hours:	F 11:30-1:30 Raven 105	X-hour:	Th 1:20–2:10
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Objectives: This course will cover the classical foundations of cognitive science, as well as the more recent developments in the field. We will start with a historical look at the cognitive sciences, then will explore philosophical views about the relationship between the mind and the physical world, and the computational theory of mind. In the second part of the course we will examine alternative approaches to mind, including connectionist views and theories of embodiment and dynamical systems. The last part of the course, if we have time, will touch on more recent developments in cognitive science, including Bayesian approaches and reinforcement learning.

Course materials: The readings from this class will be taken from the following textbook and anthology of classic papers supplemented with additional articles. I will provide PDFs of all supplemental course materials. Since we will be reading from them most weeks, you should consider purchasing the books:

Clark, Andy (2001). *Mindware: An Introduction to the Philosophy of Cognitive Science*. New York: Oxford University Press USA.

Cummins R. and D.D. Cummins (2000) *Minds, Brains, and Computers: The foundations of cognitive science*. Blackwell.

Grading:

- Presentations, quizzes and other assignments (10%)
- 2 response papers ($\approx$ 500 words) (10%)
- Midterm (15%)
- 1 short paper (4-6 pp double-spaced) (25%)
- Final paper or project (8-10 pages for an individual; 12-15 page for a group) (25%)
- Attendance and class participation: (15%)

Response papers should focus on one point in the material that you would like to respond to or expand upon. Responses could take either a critical perspective, laying out reasons for doubting the arguments made in the material, and the evidence in support of your counterargument. Both theoretical and empirical evidence is fair game. Responses can also take a more positive perspective, expanding upon the arguments

presented in the material and extending them either theoretically or empirically. **These should not be summaries** of the material you are responding to.

Short papers, like the response papers, can critique or expand upon a perspective have encountered in the course material. These short papers should include a brief summary of the material you are responding to, but should primarily focus on your own novel contribution and anticipate the best responses to the new arguments you are putting forward. If you wish, you may also propose experiments that you believe will help elucidate the answer to the theoretical question you are focusing on.

Group projects will either consist of a review of an area of cog sci not explicitly covered in the first part of the course, or a grant proposal for a novel experiment in cog sci. Groups should consist of 4-5 people. These topics should be cleared with me before proceeding.

Important Dates:

Short paper dueFebruary 6th, 2026

Final paper or project dueMarch 17th, 2026

Note: We may have x-hour meetings during the term (Wednesday). If we do, they will be announced on Canvas. You are expected to attend. If you expect this to be a problem, please talk to me.

Academic Honesty: The faculty, administration, and students of Dartmouth College acknowledge the responsibility to maintain and perpetuate the principle of academic honor, and recognize that any instance of academic dishonesty is considered a violation of the [Academic Honor Principle](#).

Religious Accommodations: Dartmouth has a deep commitment to support students' religious observances and diverse faith practices. Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible—before the end of the second week of the term at the latest—to discuss appropriate course adjustments.

Disability Accommodations: Students requesting disability-related accommodations and services for this course are required to register with Student Accessibility Services (SAS; [Apply for Services](#); student.accessibility.services@dartmouth.edu; 1-603-646-9900) and to request that an accommodation email be sent to me in advance of the need for an accommodation. Then, students should schedule a follow-up meeting with me to determine relevant details such as what role SAS or its [Testing Center](#) may play in accommodation implementation. This process works best for everyone when completed as early in the quarter as possible. If students have questions about whether they are eligible for accommodations or have concerns about the implementation of their accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Mental Health Resources: The academic environment is challenging, our terms are intensive, and classes are not the only demanding part of your life. There are a number of resources available to you on campus to support your wellness, including: the [Counseling Center](#) which allows you to book triage appointments online, the [Student Wellness Center](#) which offers wellness check-ins, and your [undergraduate dean](#). The student-led [Dartmouth Student Mental Health Union](#) and their peer support program may be helpful if you would like to speak to a trained fellow student support listener. If you need immediate assistance, please

contact the counselor on-call at (603) 646-9442 at any time. Please make me aware of anything that will hinder your success in this course.

Sexual and Gender-Based Misconduct: At Dartmouth, we value integrity, responsibility, and respect for the rights and interests of others, all central to our Principles of Community. We are dedicated to establishing and maintaining a safe and inclusive campus where all community members have equal access to Dartmouth's educational and employment opportunities. We strive to promote an environment of sexual respect, safety, and well-being. Through the Sexual and Gender-Based Misconduct Policy (SMP), Dartmouth demonstrates that sex and gender-based discrimination, sex and gender-based harassment, sexual assault, dating violence, domestic violence, stalking, etc., are not tolerated in our community.

For more information regarding Title IX and to access helpful resources, visit Title IX's website (sexual-respect.dartmouth.edu). As a faculty member, I am required to share disclosures of sexual or gender-based misconduct with the Title IX office.

If you have any questions or want to explore support and assistance, please contact the Title IX office at 603-646-0922 or TitleIX@dartmouth.edu. Speaking to Title IX does not automatically initiate a college resolution. Instead, much of their work is around providing supportive measures to ensure you can continue to engage in Dartmouth's programs and activities.

Financial Hardship Resources: If you encounter financial challenges related to this class, there may be sources of support for you. If you feel comfortable sharing your experience with me, you may. You may also consider meeting with a financial aid officer to discuss options, reaching out to the First-Generation Office if you are a first-generation student, browsing the [Funding Resources](#) page, or, for unexpected expenses, applying to the Barrier Removal Fund through the Financial Aid tile in [DartHub](#).

Tentative Course Outline:

The *italicized readings* are **optional**; everything else is required.

Week 1: introduction

- Appendix & Chapter 1 of Clark.
- MBC 5: Marr, "Vision"

Week 2: classical architectures

- Chapter 2 of Clark.
- MBC 10: Turing, "Computing Machinery and Intelligence"
- MBC 2: Haugeland, "Semantic Engines"
- CANVAS: *Newell & Simon, "Computer Science as Empirical Inquiry"*
- MBC 9: Searle, "Minds, Brains, and Programs"

Week 3: status of folk psychology

- Chapter 3 of Clark.
- MBC 4: Fodor, "The Language of Thought"
- CANVAS: Dennett, "True Believers"
- MBC 33: Churchland, "Eliminative Materialism and the Propositional Attitudes"

Week 4: connectionism

- Chapter 4 of Clark.
- MBC 12: Churchland, "Cognitive Activity in Artificial Neural Networks"
- CANVAS: Fodor & Pylyshyn, "Connectionism and cognitive architecture"
- MBC 14: *Rumelhart & McClelland, "On Learning the Past Tenses of English Verbs"*
- MBC 18: *Pinker & Prince, "Rules and Connections in Human Language"*

Week 5: doing without representation and computation?

- Chapters 5-7 of Clark.
- MBC 26: Sejnowski, Churchland & Koch, "Computational Neuroscience"
- CANVAS: Brooks, "Intelligence without representation"
- CANVAS: *van Gelder, "What might cognition be, if not computation?"*

Week 6: 4E theories of cognition

- Chapters 9-10 of Clark.
- CANVAS: Clark & Chalmers, "The extended mind"
- CANVAS: Varela, Thompson, & Rosch, "The Embodied Mind"
- CANVAS: *O'Regan & Noë, "A sensorimotor account of vision and visual consciousness"*

Week 7: nativism vs. empiricism

- MBC 28: Chomsky, “Recent contributions to the theory of innate ideas”
- MBC 31: Spelke, “Initial knowledge: six suggestions”
- MBC 29: Putnam, “*The ‘innateness hypothesis’ and explanatory models in linguistics*”
- CANVAS: Pullum & Scholz, “*Empirical assessment of stimulus poverty arguments*”
- CANVAS: Crain & Pietroski, “*Why language acquisition is a snap*”

Week 8: Bayesian cognitive science and predictive processing

- Chapter 11 of Clark.
- CANVAS: Tenenbaum, Kemp, Griffiths & Goodman, “How to grow a mind: Statistics, structure, and abstraction”
- CANVAS: Clark, “Whatever next? Predictive brains, situated agents, and the future of cognitive science”

Week 9: reinforcement learning

- CANVAS: Botvinick et al., “Reinforcement Learning, Fast and Slow”